Wildfire Insured-Loss Modelling

Frequency-Severity Loss Model for an Alberta Property Portfolio,

Projected Under a Future Climate Scenario (EC-Earth3, SSP1-2.6)

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2026

Objective & Approach

Goal
Quantify wildfire insurance risk by FSA and portfolio-wide, and project losses under a future climate scenario.

Framework
Frequency-severity structure — Negative Binomial GLM for claim frequency, Gamma GLM for claim severity.

Features
Climate (Fire Weather Index) + geospatial (fuel type) + exposure, combined into one claims feature table.

Discipline
Every non-obvious result was independently verified against source data, not assumed — model complexity was deliberately scoped to what 2 historical events can support.

Scope
A transparent, defensible baseline — not a production catastrophe model.

Project Workflow

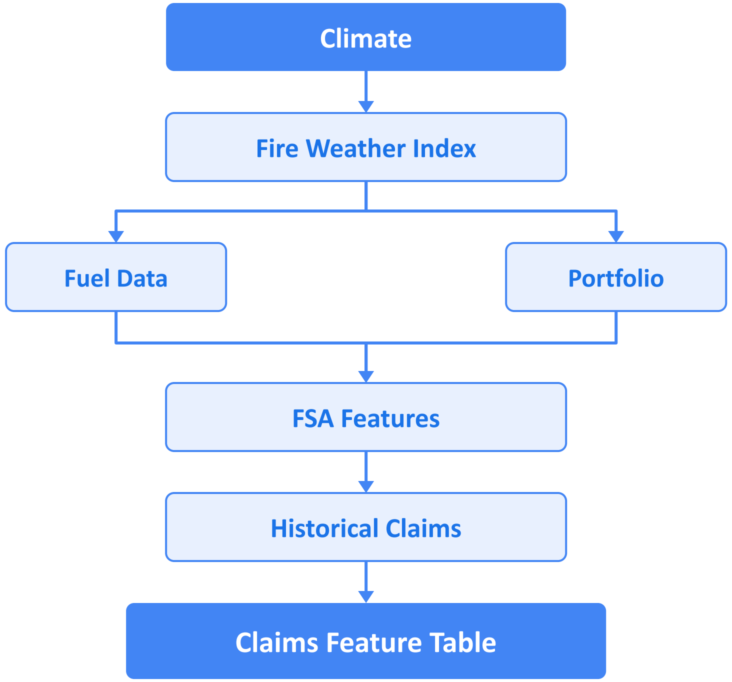
- Raw claims + climate + fuel data
- Feature engineering
- Poisson baseline
- Overdispersion → Negative Binomial
- Multiple FWI features → Multicollinearity
- Feature reduction (2 climate + 1 fuel)
- Validation & diagnostics (LOEO + spatial dependence)
- Future climate scenario (SSP1-2.6)
- **Figures • Maps • Results • Report**

Data Sources & Pipeline

Data Sources

- **Portfolio**
  159 FSAs (Alberta region)
- **Claims**
  305 rows; 2 wildfire events
- **CMIP6 Climate**
  Global climate projections
- **Canadian Fuel Raster**
  National fuel type mapping
- **StatCan FSA Boundaries**
  Official spatial polygons

Data Pipeline



Validation

Key assumptions regarding climate, spatial data, and joins have been verified and corrected.

Testing

High-risk transformations are covered by pytest, which also successfully identified a minor data artifact.

Boundary Gaps

Five FSAs are missing polygons, including two valid FSAs (**T3T** and **T4K**) that require resolution.

Exploratory Insights

Losses are concentrated

Most FSA/event rows record zero loss; nonzero losses span ~6 orders of magnitude ($62 to $33.9M).

Severity ≠ exposure scaling

No clear linear relationship between exposure value and loss — whether an FSA was affected matters more than its size.

24% of claim rows lack FWI signal

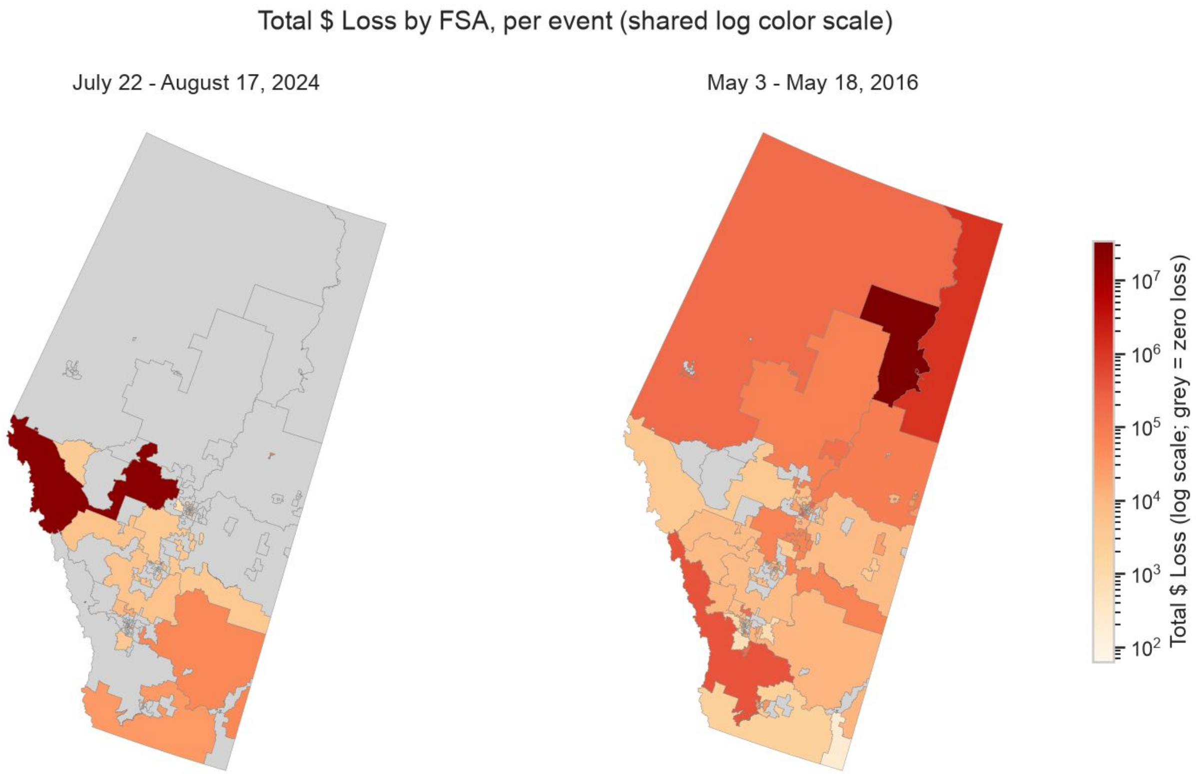
Investigated, not discarded — mostly a genuine pre-fire-season timing issue in the May 2016 event.

Spatial signatures differ by event

May 2016 losses are geographically clustered (one spreading fire); Jul/Aug 2024 losses are scattered (multiple disconnected fires).

FWI distributions are multi-modal

Consistent with the two events drawing from different fire-weather regimes, not one smooth distribution.



Feature Engineering & Model Evolution

Feature Selection Logic

- **Climate:**fwi_max (headline fire-weather hazard), dc_max (drought/fuel-dryness proxy, r≈0.35 with fwi_max).
- **Geospatial:**dominant_fuel_pct — low collinearity with both climate features (r<0.1).
- **Constraint:**Why only 3 predictors? FWI-family features are almost collinear (r>0.9) — EDA ruled out using several together.

Model Evolution Narrative

1. **Poisson baseline** → badly overdispersed.
2. **Negative Binomial** → substantially better fit.
3. **Multiple FWI features together** → unacceptable multicollinearity.
4. **Final specification** → 2 climate predictors + 1 geospatial predictor, followed by validation, spatial-dependence testing, and future-scenario application.

Model Evolution Pipeline

1. Poisson baseline GLM
2. Overdispersion identified
3. Negative Binomial GLM
4. Multicollinearity identified
5. Feature Selection (2 Climate + 1 Fuel)
6. Model Validation (LOEO + Spatial)
7. Future Climate (SSP1-2.6 Illustrative)
8. Final Robust Model

Model Results — Frequency & Severity

Frequency model	AIC
Poisson	135,998.1
Negative Binomial	**1,196.8**

Performance

Predicted loss: 39% of actual

MAE: $573K

RMSE: $3.34M

Frequency Model (Negative Binomial)

Geospatial feature earns its place:

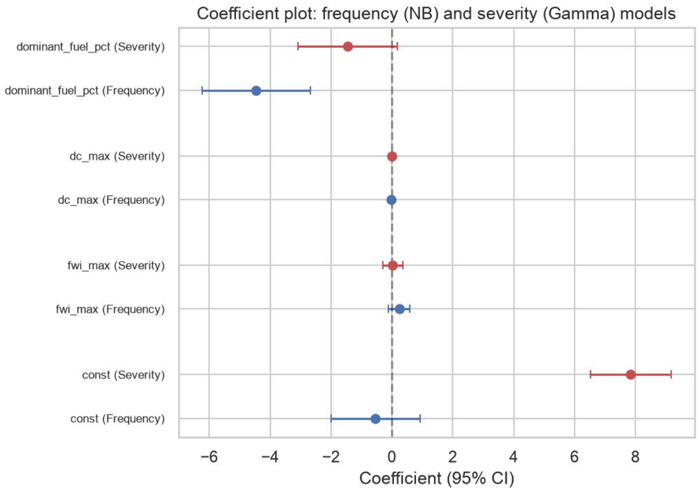
Adding **`dominant_fuel_pct`** improves AIC by 21.0 and a likelihood-ratio test rejects the reduced (climate-only) model strongly ($p=1.65\times10^{-6}$).

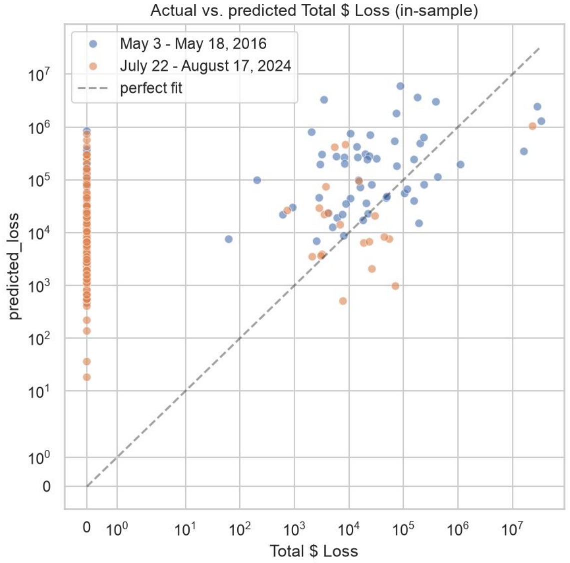
Severity Model (Gamma)

Limited statistical power:

No predictor is significant on 76 usable rows — indicating limited power to detect relationships rather than confirming that none exist.

Model Coefficients & Prediction Alignment





Metric	Value
Predicted / actual total loss	$41.4M / $106.9M (39%)
MAE	$572,725
RMSE	$3,336,833

Validation — Out-of-Sample & Spatial Dependence

Leave-One-Event-Out (LOEO)

Train → Test	Predicted	Actual	Rank corr.
May 2016 → Jul/Aug 2024	$186K	$23.7M	≈0 (0.068)
Jul/Aug 2024 → May 2016	$189.9M	$83.2M	0.297

Unreliable

Insufficient sample size

- Two wildfire events are entirely insufficient for a trustworthy, robust out-of-sample validation of the model.

Spatial Dependence (GEE)

≈ 0

(-0.008) estimated within-event correlation

- Not a precise estimate. With only 2 clusters, this metric cannot reliably demonstrate whether spatial dependence exists.

Bottom Line:

With only two wildfire events, neither out-of-sample performance nor spatial dependence can be estimated reliably. This is not evidence that either is absent, but rather that the currently available data cannot yet answer the question.

Future Climate Projection (Illustrative)

Labelled "illustrative," not "predicted"
Model trained on event-window extremes; future comparison uses period-average inputs — a feature-scale mismatch that pushes it outside training distribution.

No prediction interval reported
Parameter uncertainty from only 2 historical wildfire events would make any prediction interval itself highly unstable and unreliable.

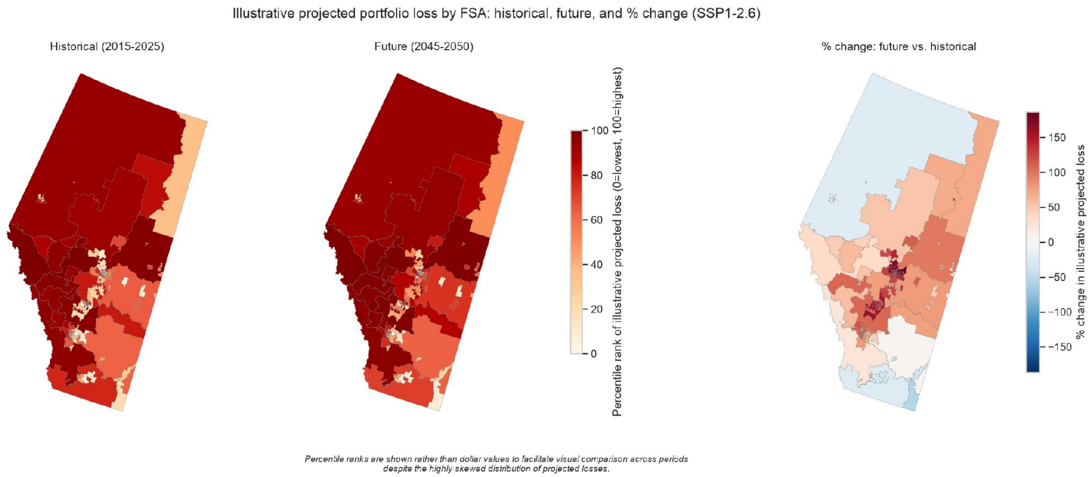
Single model, scenario, & ensemble member
A narrow projection pathway, whereas other CMIP6 climate models often project overall increasing fire-weather risk for similar geographic regions.

Frequency/loss rise despite FWI falling
Driven by a negative dc_max coefficient learned from only 2 events — which is plausibly a statistical artifact rather than a stable physical relationship.

Metric	Historical (2015–2025)	Future SSP1-2.6 (2045–2050)	Rel. diff.
Mean FWI	3.47	1.36	−61%
Mean DC	203.6	148.7	−27%
Illustrative claim frequency	1,870	3,862	+107%
Illustrative projected loss	$4.5M	$7.65M	+70%

Illustrative Projected Portfolio Loss by FSA

Historical, future, and % change (SSP1-2.6)



Percentile ranks are shown rather than dollar values to facilitate visual comparison across periods despite the highly skewed distribution of projected losses.

Limitations, Next Steps

Key limitations:

- **Only 2 historical events.** Thin basis for frequency/severity or tail-risk estimation.
- **Poor out-of-sample generalization.** LOEO failed to converge in both directions.
- **Spatial dependence imprecisely estimated.** Only 2 clusters.
- **Open data gaps.** 24% of claims lack FWI signal; T3T/T4K boundary gap; fuel codes unlabelled.
- **Single climate pathway.** One model, one scenario, one ensemble member.

Next steps (by business value):

1. **More historical events.** The single highest-value addition.
2. **Monte Carlo simulation.** Real prediction intervals once enough events exist.
3. **Multi-model climate ensemble.** Beyond one model/scenario/member.
4. **Resolve remaining data gaps.** FWI coverage, boundary gap, fuel legend.
5. **Alternative severity predictors.** An exposure-oriented set, not climate-only.

Conclusion: the modelling framework is technically sound and reproducible, but additional historical wildfire events are the critical requirement before operational deployment.